

TestBoard 7953 — channel-11 test results

9 October 2026 · Both sensor strips unplugged · Laptop charger connected

ADC	Vmid bias resistor	Channels routed to connector	PZT groups on connector
ADC1	1 MΩ	0–9	PZT6, PZT7
ADC2	470 kΩ	0–14	PZT1, PZT3, PZT5
ADC3	470 kΩ	0–9	PZT6, PZT7
ADC4	249 kΩ	0–14	PZT1, PZT3, PZT5

This table describes the board wiring; both sensor strips were unplugged during these tests. Each bias resistor connects Vmid to the shared MXO / OPA365 input, rather than to individual channels. Channel 15 is the Vmid reference on every ADC.

The measured reference stays near 1250 mV while channel-11 baselines depend on SPI speed and selection history. Vmid insertion lowers channel 11 in every single-ADC hold pair, including ADC1 and ADC3. This supports a dynamic analog-path explanation; it does not identify the exact mechanism.

With Vmid off, channel 11 is below its neighbors on ADC2/ADC4, but slightly above them on ADC1/ADC3. With Vmid on, it is below its neighbors on all four ADCs.

40 captures, 39,615,388 received sweeps. Production firmware restored and stopped; COM3 closed. All graphs use nominal mV = raw code × 2500/4096.

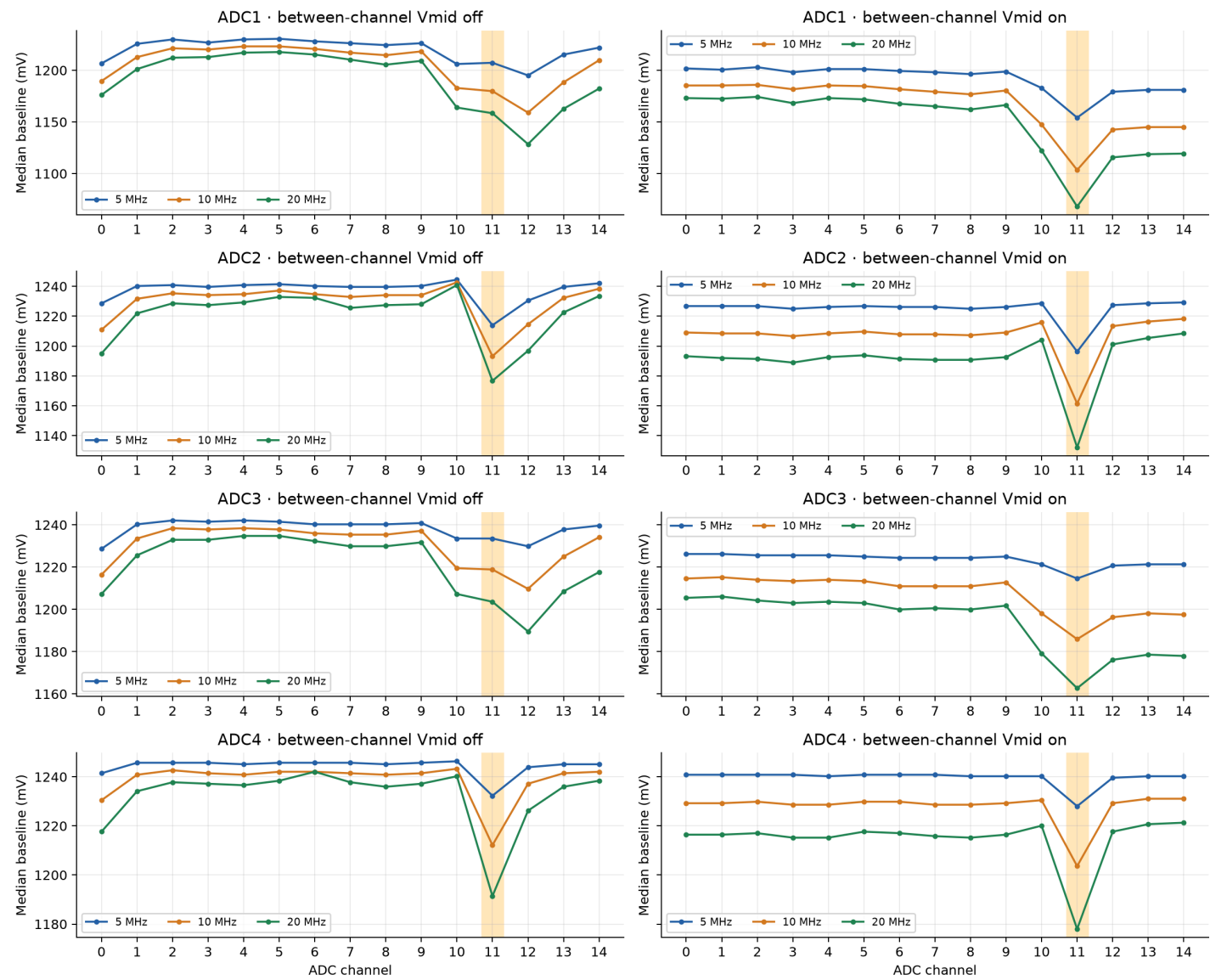
Channel 11 at 10 MHz (mV)

ADC	Scan, Vmid off	Scan, Vmid on	Hold, Vmid off	Hold, Vmid on
ADC1	1179.81	1103.52	1166.99	1049.80
ADC2	1193.24	1161.50	1220.70	1118.16
ADC3	1218.87	1185.91	1206.05	1149.90
ADC4	1212.16	1203.61	1229.25	1161.50

SPI speeds: all channels

Each ADC, channels 0–14, at 5/10/20 MHz; Vmid off and on.

SPI speed comparison — all four ADCs, channels 0–14

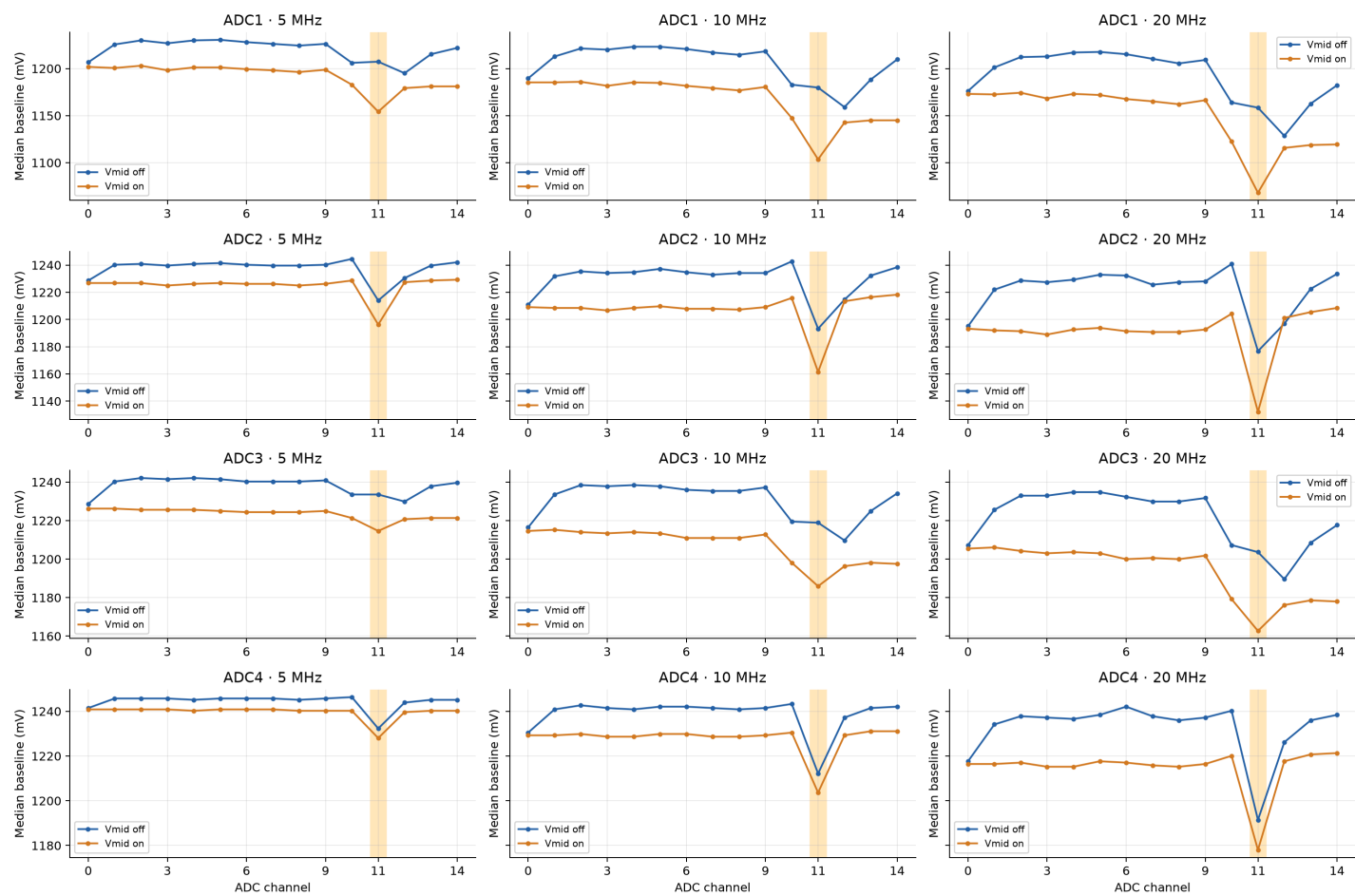


Exact 60-route scans; channel 11 highlighted. 12 s after 1 s warm-up; no connected sensor strips.

Vmid off versus on

The same clock and routes; each ADC plotted separately.

Effect of Vmid between channels — baseline versus channel

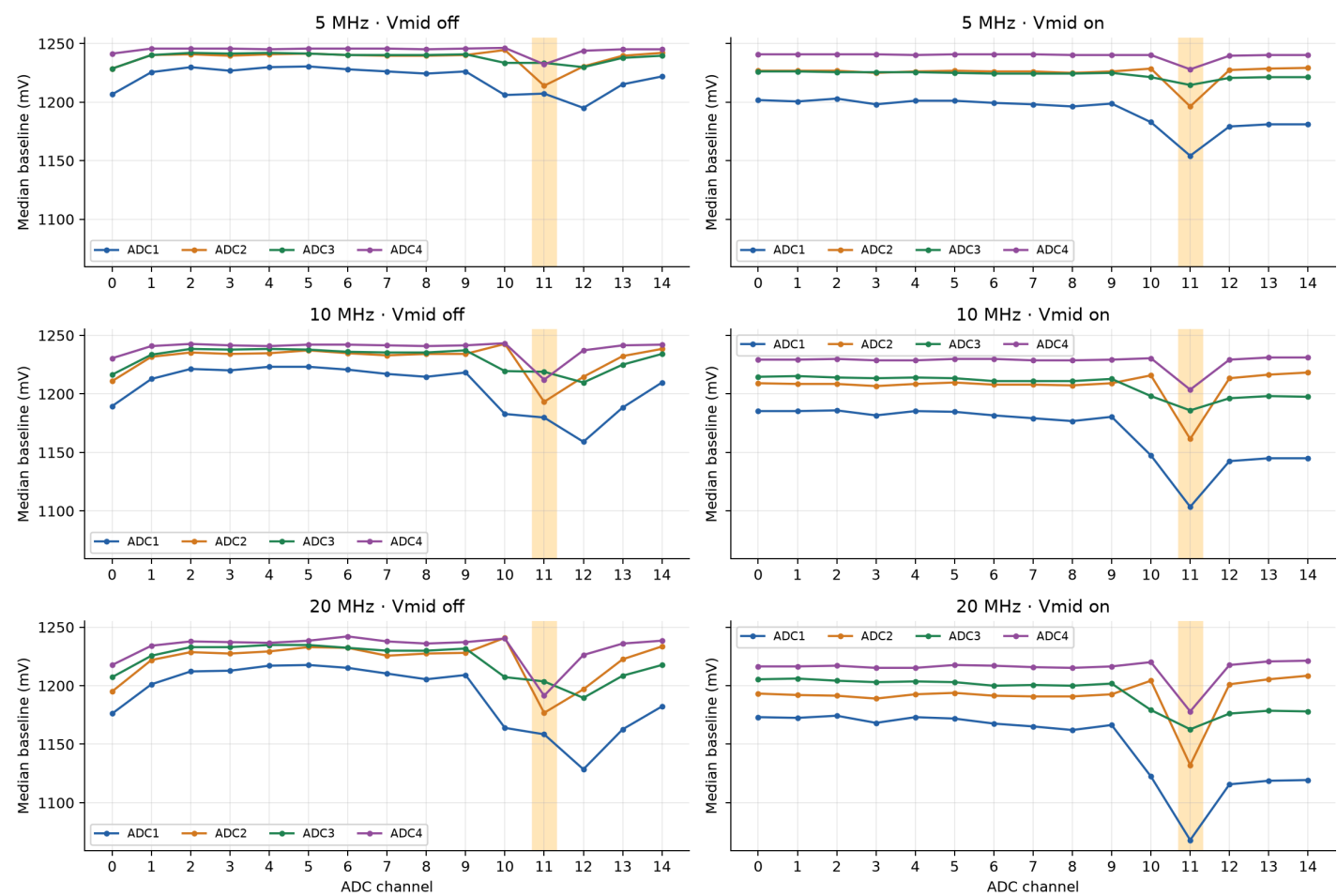


Exact 60-route scans; same manual/blocking/repeat1 settings. mV = raw code × 2500/4096.

Four ADCs at the same SPI speed

All four ADCs overlaid, including previously unused channels 10–14 on ADC1/ADC3.

Four ADCs at the same SPI speed

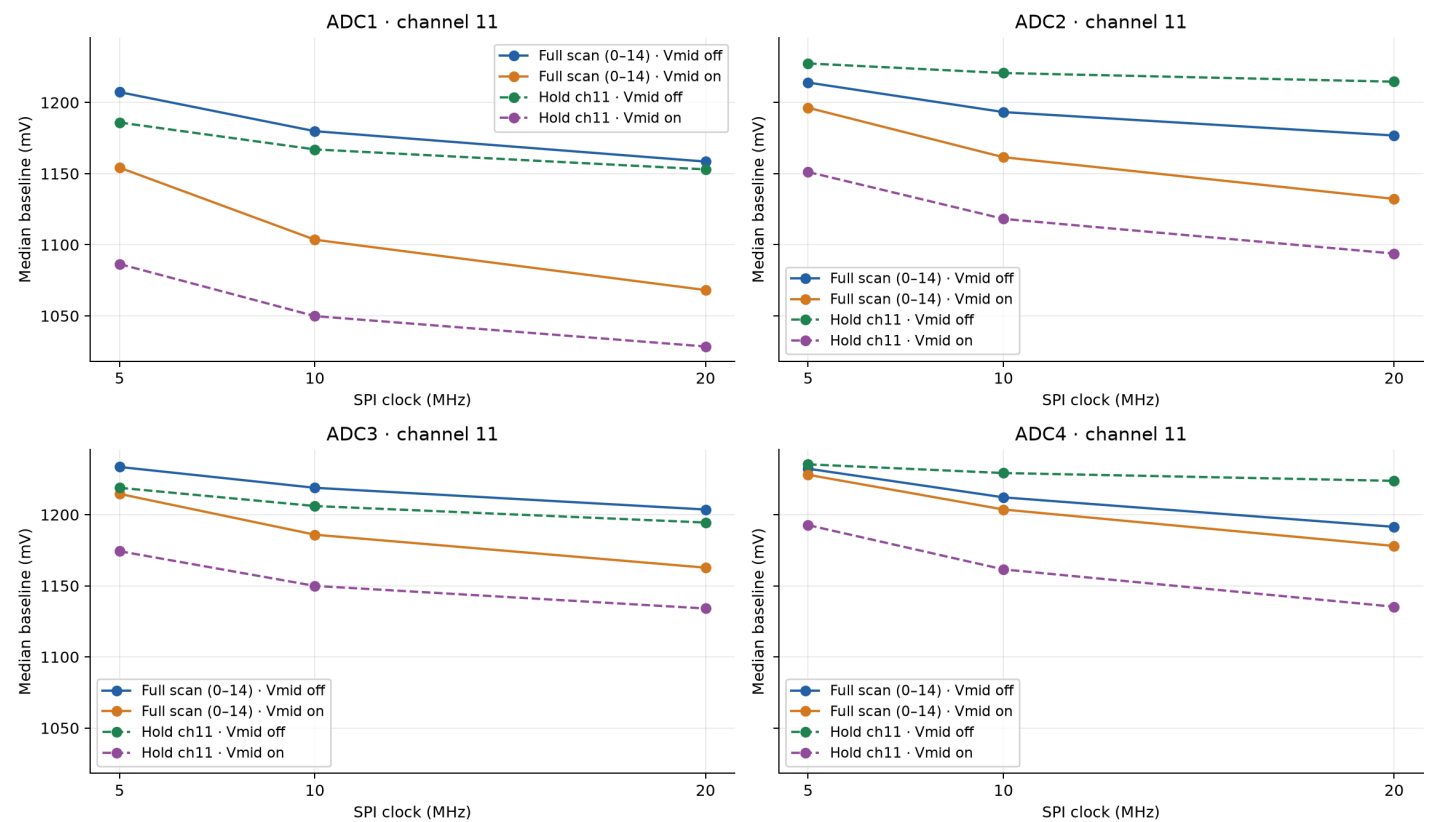


All four ADCs captured together, each with channels 0–14. Channel 11 highlighted; medians in mV.

Channel 11: full scanning versus continuous selection

Full scan: all four ADCs scan channels 0–14, with only channel 11 shown here. Hold: one ADC alone selects channel 11 repeatedly; with Vmid on, it repeats 11 → 15 → 15 and retains only channel 11. At 10 MHz, a full scan produces one channel-11 result per ADC about every 159 μ s with Vmid off or 415 μ s with Vmid on. Either single-ADC hold mode produces a retained channel-11 result about every 8 μ s. Comparing a scan with a hold therefore changes both selection history and sampling cadence.

Channel 11 — scanning versus holding one ADC alone

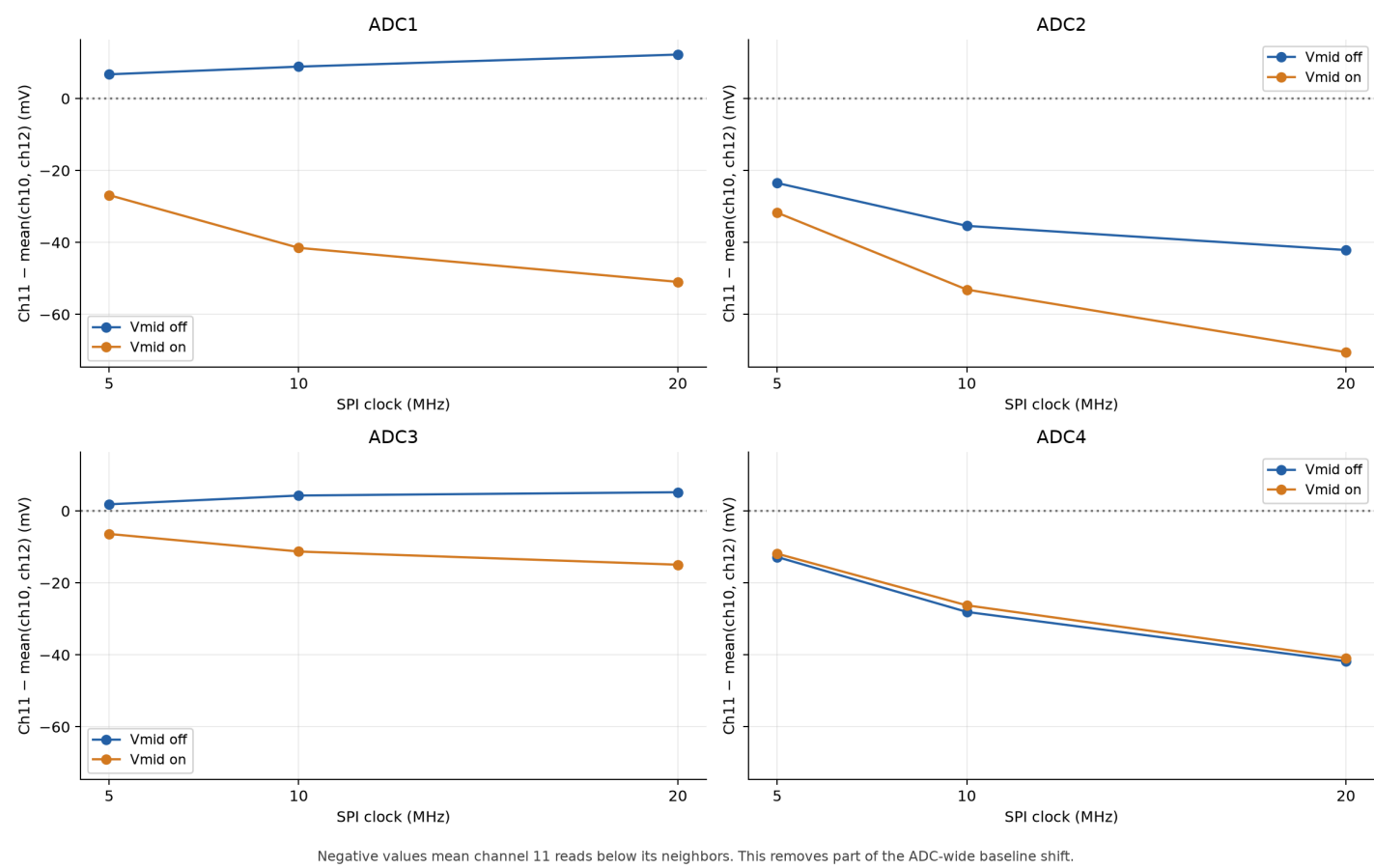


Full scan: all 4 ADCs, channels 0–14; curves show only ch11. Hold: one ADC alone selects ch11 repeatedly. Vmid-on hold repeats 11 → 15 → 15; only ch11 retained. At 10 MHz: scan off/on 159/415 μ s; either hold 8 μ s.

Channel 11 relative to channels 10 and 12

Negative values show a channel-11 deficit after removing the neighboring baseline level.

Channel 11 relative to its neighbors — full 60-route scans

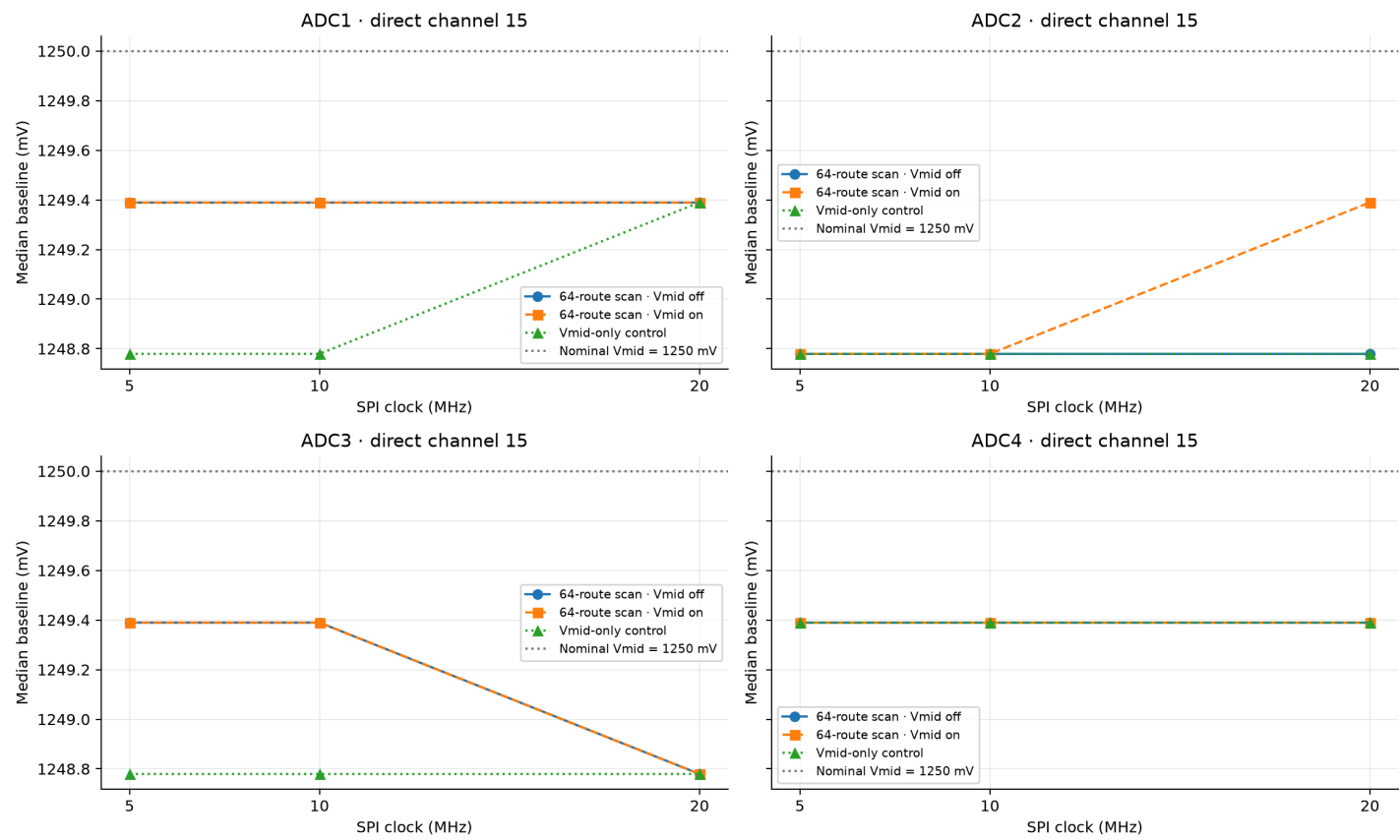


Negative values mean channel 11 reads below its neighbors. This removes part of the ADC-wide baseline shift.

Measured Vmid / channel 15

Companion full scans versus separate reference-only controls.

Measured Vmid / channel 15 — all four ADCs

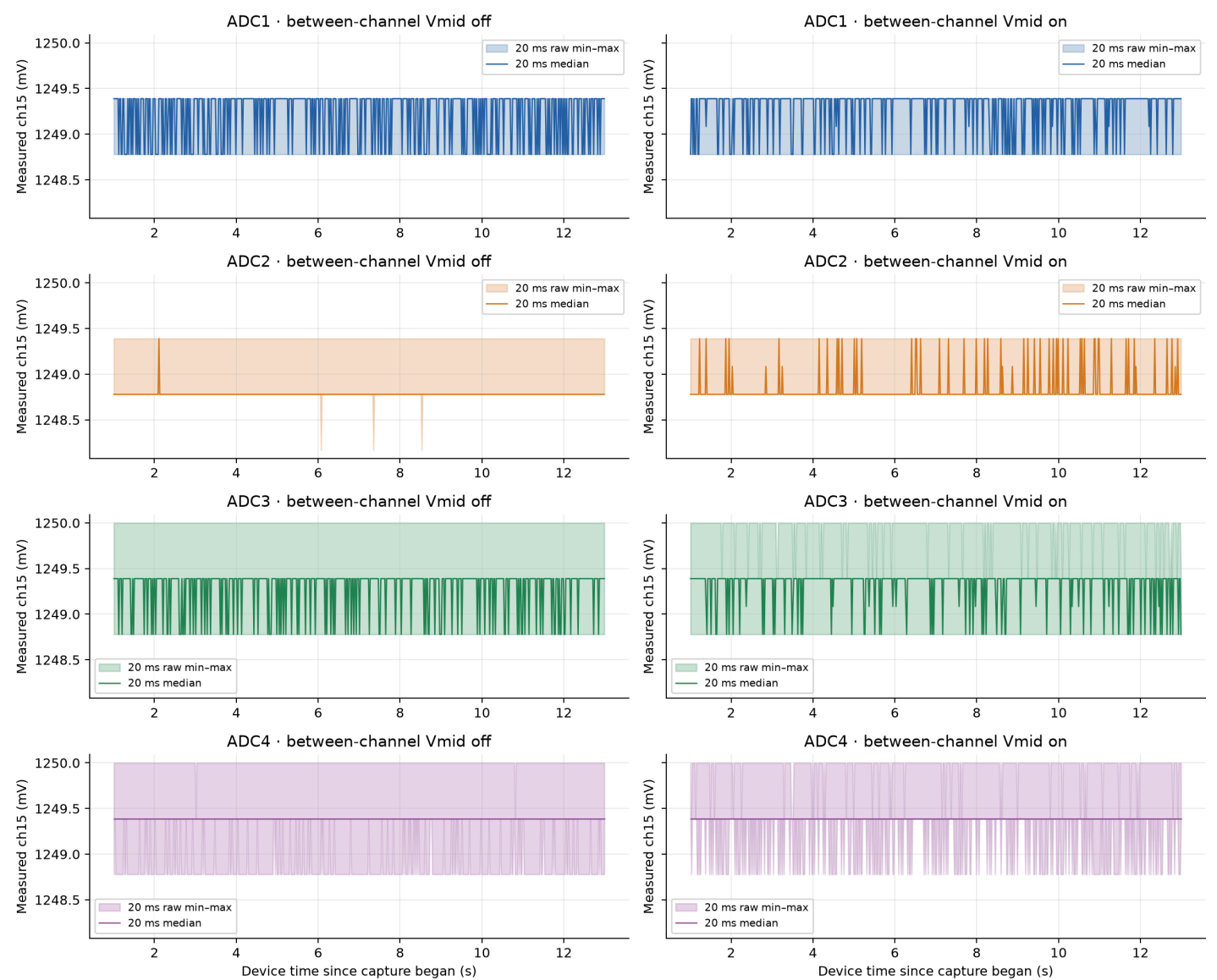


Companion 64-route scans retain ch15 after ch14; reference-only controls are separate captures, not derived estimates.

Vmid time traces

10 MHz examples, all ADCs, Vmid off/on; every raw sample contributes to each 20 ms min-max envelope.

Vmid time traces at 10 MHz — every sample retained in the envelope



Direct reference conversions from companion 64-route scans. Shaded envelope includes all samples; no interpolation or noise filtering.